

y-direction: $0 = p_x \sin\theta - p_y \sin\alpha$

8 D Extension $x = F/k$

Both string are stretched by same force 80 N.

$$x_1 = 80/4000 = 0.02\text{m}, x_2 = 80/2000 = 0.04\text{m}$$

$$\text{ratio of pe} = \frac{1}{2} Fx_1 / \frac{1}{2} Fx_2 = x_1 / x_2 = 1:2$$

9 A During 2.0 s, displacement $s = \text{area under v-t graph} = \frac{1}{2} \times 2.0 \times 0.5 = 0.5 \text{ m}$